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Metacognitive Awareness Habitation in Primary Students' Speaking Practices through Digital Microlearning

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Abstract

Background/purpose. Primary-level speaking instruction largely prioritizes fluency and accuracy, offering limited opportunities for learners to regulate speaking through planning, monitoring, and reflection. While metacognitive awareness has been extensively examined in reading and writing, empirical evidence on its habituation in primary students' speaking remains scarce. This study aimed to examine how metacognitive awareness is systematically habituated in primary students' speaking practices through digital microlearning as a routine classroom scaffold.

Materials/methods. A classroom-based, process-oriented intervention was conducted using a single-group descriptive pre-post design with qualitative triangulation. Twenty-eight Grade 5 students participated in eight speaking sessions over four weeks. Digital microlearning prompts were consistently embedded before and after speaking tasks to support planning, monitoring, and reflection. Data were collected through observation rubrics, student reflections, and teacher observation notes, and analyzed descriptively and thematically.

Results. The findings indicated a gradual and consistent increase in observable metacognitive speaking behaviors. Overall metacognitive awareness scores improved from 1.32 to 2.64, with clear gains in planning, monitoring during speech, and post-speaking reflection.

Qualitative evidence showed a shift from surface-level evaluation toward process-oriented self-regulation.

Conclusion. The study demonstrates that metacognitive awareness in primary students' speaking can be effectively habituated through repeated reflective routines supported by digital microlearning, extending metacognitive research beyond literacy skills and positioning speaking as a viable domain for metacognitive regulation in primary education.

1. Introduction

Speaking is a foundational component of language development in primary education, functioning not only as a means of communication but also as a medium through which children construct meaning, organize ideas, and demonstrate understanding. At the elementary level, speaking activities ideally support the development of oral fluency, coherence, and communicative confidence while simultaneously fostering higher-order cognitive engagement. However, research across diverse educational contexts indicates that speaking instruction in primary

classrooms often remains limited to surface-level performance, emphasizing correctness and fluency rather than reflective language use (Goh & Burns, 2012).

In the Indonesian primary education context, speaking instruction tends to be strongly teacher-directed and product-oriented. Classroom speaking activities frequently take the form of answering factual questions, retelling texts, or reproducing model sentences provided by teachers or textbooks. While such practices may support basic oral production, they rarely encourage learners to reflect on what they intend to say, how they express their ideas, or how effectively their message is conveyed to others. As a result, speaking is commonly treated as an output activity rather than a cognitive and reflective process (Santiani & Putra, 2021). This condition mirrors broader findings in language education research showing that young learners' oral language development is often constrained by instructional routines that prioritize reproduction and memorization over meaning-making and self-regulation (Goh & Liu, 2023).

The limited integration of metacognitive processes further compounds this issue. Metacognitive awareness refers to learners' consciousness of their own thinking processes and their ability to plan, monitor, and evaluate learning activities (Flavell, 1979). In speaking, metacognitive awareness enables learners to anticipate content before speaking, monitor clarity and coherence during speech, and reflect on performance afterward. These processes are particularly important for young learners, as early exposure to metacognitive practices contributes to the gradual development of self-regulated learning behaviours (Whitebread et al., 2009). Despite this, metacognitive instruction in primary education has predominantly focused on reading and writing, while speaking remains underexplored as a domain for metacognitive development, especially in the Indonesian context.

Existing scholarship has repeatedly shown that speaking instruction in primary classrooms remains predominantly reproductive and procedural in nature. Mercer (2013) and Goh and Burns (2012) note that classroom talk is often structured to elicit brief, teacher-expected responses, positioning students as language reproducers rather than meaning-makers. In such contexts, learners are seldom encouraged to engage in reflective processes before speaking, to monitor the coherence and clarity of their utterances while speaking, or to evaluate the effectiveness of their communication afterward (Mercer & Dawes, 2014; Zhang & Zhang, 2019). Studies examining metacognition in oral language learning further suggest that feedback practices in speaking lessons are largely external and evaluative, emphasizing accuracy and fluency over reflective self-regulation (Whitebread et al., 2009; Andrade & Brookhart, 2020). Therefore, opportunities for learners to internalize evaluative criteria and develop awareness of their own speaking processes remain limited. Goh and Liu (2023) argue that when reflective support is absent, speaking activities become disconnected from conscious cognitive regulation, reducing students' capacity to plan, monitor, and adapt their oral language use across contexts.

Taken together, these findings reveal a persistent mismatch between contemporary calls for meaningful, reflective learning and the realities of speaking

pedagogy in primary education. While theoretical and policy discourses increasingly emphasize learner awareness, self-regulation, and deep learning (Kementerian Pendidikan Dasar dan Menengah Republik Indonesia, 2025), everyday classroom practices continue to privilege observable performance over the invisible metacognitive processes that underpin effective and sustainable speaking development.

To address this limitation, the present study is grounded in the concept of habituation as a pedagogical process. Habituation refers to the formation of stable cognitive and behavioral patterns through repeated, consistent practice in meaningful contexts (Lally et al., 2010). From a developmental perspective, habituation is especially relevant for primary school learners, as young children are more likely to internalize cognitive routines through simple, repetitive activities rather than abstract instruction. In this study, habituation is conceptualized as the gradual embedding of metacognitive awareness into students' speaking practices through recurring reflective routines.

Table 1 presents the theoretical framework of habituation and its operationalization within the context of this study.

Aspect	Description in Habituation	Application
Repetition	Habits develop through repeated actions	Regular reflective speaking activities
Consistency	Stable routines support internalization	Fixed reflection prompts across sessions
Simplicity	Tasks must match learners' developmental level	Child-friendly metacognitive questions
Meaningful context	Habits form in relevant situations	Speaking topics drawn from students' daily experiences
Internalization	External guidance becomes self-initiated	Students gradually reflect without prompts

Within this framework, metacognitive awareness in speaking is not taught as an explicit concept but cultivated as a habitual practice embedded in routine classroom activities. This approach aligns with calls for pedagogical designs that support deep and meaningful learning while remaining developmentally

appropriate for primary students (Kementerian Pendidikan Dasar dan Menengah Republik Indonesia, 2025).

Digital microlearning is positioned in this study as a supportive learning context rather than a technological focus. Microlearning, characterized by short, focused learning units, aligns well with the attention span and learning preferences of primary students and has gained popularity in elementary classrooms as a flexible digital medium (Leong et al., 2021). When used strategically, digital microlearning can scaffold brief reflective moments before, during, and after speaking activities, thereby supporting the habituation of metacognitive awareness without overburdening instructional time or teacher workload.

Despite growing interest in meaningful learning and digital integration in primary education, a clear gap remains between these aspirations and actual classroom practices in speaking instruction. Current approaches rarely combine speaking, metacognitive habituation, and realistic digital support in a coherent pedagogical design. This gap underscores the need for empirical research that examines how metacognitive awareness in speaking can be habituated through feasible classroom interventions.

Accordingly, this study seeks to examine how metacognitive awareness is habituated in primary students' speaking through digital microlearning as a routine classroom practice. Rather than focusing on short-term performance gains, the study adopts a process-oriented perspective by investigating the formation of reflective speaking habits over repeated learning cycles. Specifically, this study addresses the following research questions: (1) how metacognitive awareness is habituated in primary students' speaking practices through digital microlearning; (2) what observable behavioural indicators reflect the development of metacognitive awareness in students' speaking; and (3) how primary students respond to the use of digital microlearning in supporting reflective speaking routines. By answering these questions, the study aims to bridge the gap between the demand for meaningful, reflective learning and the prevailing reproductive practices of speaking instruction in primary education.

2. Literature Review

Previous research has consistently shown that speaking instruction in primary education tends to prioritize observable performance outcomes such as fluency, pronunciation, and accuracy, while offering limited support for learners' reflective regulation of oral production (Goh & Burns, 2012; Mercer, 2013). Studies on classroom talk indicate that speaking activities are often structured to elicit brief, teacher-expected responses, positioning students as language reproducers rather than active meaning-makers (Mercer & Dawes, 2014; Zhang & Zhang, 2019). In contrast, metacognitive awareness, defined as learners' ability to plan, monitor, and evaluate their own learning processes, has been widely recognized as a key component of self-regulated learning and sustained language development (Flavell, 1979; Veenman et al., 2006). Empirical research in language education has demonstrated the

effectiveness of metacognitive instruction in enhancing reading and writing outcomes across age groups (Whitebread et al., 2009; Andrade & Brookhart, 2020). However, comparatively little attention has been given to speaking as a domain for metacognitive development, particularly at the primary level, where instructional practices remain largely product-oriented (Santiani & Putra, 2021; Goh & Liu, 2023). Recent studies suggest that metacognitive awareness in speaking is more likely to emerge when learners engage in structured reflective routines rather than receiving solely external feedback (Zhang & Zhang, 2019). Additionally, microlearning has been identified as a developmentally appropriate digital approach that supports short, focused cognitive engagement and routine formation in young learners (Leong et al., 2021). Nevertheless, existing literature rarely integrates speaking, metacognitive habituation, and digital microlearning within a coherent pedagogical design, indicating a clear gap that the present study seeks to address.

3. Methodology

This study employed a classroom-based, process-oriented intervention design to investigate the habituation of metacognitive awareness in primary students' speaking practices. The methodological orientation of the study draws on habit formation theory, which conceptualizes learning as the gradual internalization of cognitive routines through repeated and consistent practice in authentic contexts (Lally et al., 2010). In educational research, such an orientation is commonly operationalized through short-cycle classroom interventions aimed at documenting developmental processes rather than testing causal effectiveness (Whitebread et al., 2009; Cobb et al., 2003).

Accordingly, the study adopted a single-group descriptive pre–post design with embedded qualitative analysis, a design frequently used in classroom research to examine pedagogical processes that unfold over time under natural instructional conditions (Creswell & Plano Clark, 2018). This design was selected because the focus of the study was not to compare treatment effects across groups, but to examine how metacognitive awareness in speaking emerged and stabilized as a habitual practice through repeated instructional routines. Such an approach is particularly appropriate for primary education settings, where ethical, practical, and curricular constraints often limit the feasibility of controlled experimental designs.

The participants were 28 Grade 5 students enrolled in a public elementary school in Indonesia, aged between 10 and 11 years. This age group was selected based on developmental research indicating that children at this stage can emerge metacognitive awareness when supported through structured and concrete learning routines (Whitebread et al., 2009). The class was selected through convenience sampling, reflecting an intact classroom accessible to the researcher. All students participated as part of their regular classroom learning activities, and no exclusion criteria were applied, as the study aimed to examine habituation at the classroom level rather than individual differences.

The intervention was implemented over a period of four weeks, comprising eight speaking sessions conducted twice per week. This duration was determined with reference to habit formation research, which suggests that observable behavioural regularities can begin to emerge through repeated practice within a relatively short time frame, provided that instructional routines are consistent and stable (Lally et al., 2010). Each session followed an identical pedagogical structure to support habituation rather than novelty-driven engagement.

Digital microlearning was used as a pedagogical scaffold to support reflective speaking routines. The microlearning units consisted of short, focused digital prompts accessed by students immediately before and after speaking activities. These prompts were designed to elicit metacognitive awareness related to planning, monitoring, and reflection, using language appropriate for primary learners. Importantly, metacognitive concepts were not introduced explicitly; instead, students encountered the same reflective prompts across sessions, allowing awareness to develop implicitly through repeated engagement. This design choice aligns with research suggesting that young learners internalize metacognitive routines more effectively through practice than through abstract explanation (Veenman et al., 2006).

Each speaking session was organized into three pedagogical phases. Before speaking, students engaged with microlearning prompts that encouraged them to plan what they would say and how they would organize their ideas. During speaking, students performed short oral tasks, such as explaining a familiar topic or sharing personal experiences, while being reminded to attend to clarity and coherence without interrupting their speech. After speaking, students completed brief guided reflections supported by microlearning prompts, focusing on what aspects of their speaking were effective and what could be improved. The reflective structure remained constant across all sessions to reinforce habituation through repetition and consistency.

Data were collected from multiple sources to capture both observable behavioral changes and students' reflective engagement. Observable speaking behaviors were documented using a researcher-developed observation rubric focused on indicators of metacognitive awareness, including evidence of planning, monitoring, and reflection. In addition, short written reflections produced by students were collected periodically to capture their awareness of speaking processes. Teacher observation notes were also gathered to document classroom dynamics and changes in students' engagement over time. The use of multiple data sources enabled triangulation and strengthened the credibility of the findings (Miles et al., 2014).

Quantitative data from the observation rubric were analyzed descriptively to identify patterns and trends in students' metacognitive speaking behaviors across sessions. Rather than inferential testing, the analysis focused on changes in frequency and consistency of observable indicators, which is consistent with the process-oriented nature of habituation research. Qualitative data from student reflections and teacher notes were analyzed thematically using an inductive coding approach. Initial codes were generated based on core metacognitive dimensions,

namely planning, monitoring, and reflection, and subsequently refined to identify patterns indicative of habitual engagement. Integration of quantitative and qualitative findings was used to provide a coherent account of how metacognitive awareness was habituated through the intervention (Creswell & Plano Clark, 2018).



Figure 1. Cover of the Journal

Flow of the Research Procedure

Phase	Instructional Activity	Metacognitive Focus	Data Collected
Pre-intervention	Baseline observation	speaking Existing habits	speaking Observation rubric
Intervention Cycle	Repeated sessions with microlearning	speaking digital Planning, monitoring, reflection	Rubric, reflections, teacher notes
Post-intervention	Final observation and reflection	speaking Stabilization of metacognitive awareness	Rubric, reflections

4. Results

The results demonstrate a gradual increase in primary students' metacognitive awareness in speaking across the intervention period. Changes were observed in the consistency and frequency of metacognitive behaviors rather than in immediate performance gains.

Changes in Metacognitive Speaking Behaviors

Table 2 presents descriptive statistics comparing students' metacognitive speaking behaviors before and after the intervention.

Table

2

Changes in Metacognitive Speaking Behaviors (N = 28)

Metacognitive Indicator		Pre-intervention Mean	Post-intervention Mean
Planning speaking	before	1.42	2.68
Monitoring speaking	during	1.36	2.51
Reflection speaking	after	1.18	2.74
Overall metacognitive awareness		1.32	2.64

Note. Scores ranged from 1 (not observed) to 3 (consistently observed).

The data show increases across all indicators, with the largest gain observed in post-speaking reflection.

Frequency of Observable Metacognitive Behaviors

Observable speaking behaviors also changed over the course of the intervention. Table 3 presents the percentage of students demonstrating specific metacognitive behaviors during early and final sessions.

Table 3
Frequency of Observable Metacognitive Behaviors

Behavior	Early Sessions (%)	Final Sessions (%)
Pausing to plan before speaking	21	64
Self-correction during speaking	18	57
Voluntary reflection after speaking	14	71

The results indicate increased engagement in planning, monitoring, and reflection behaviors as the intervention progressed.

Qualitative Patterns in Student Reflections

Analysis of student reflections revealed changes in the nature of students’ responses. Early reflections were brief and evaluative, while later reflections increasingly referenced speaking processes. Table 4 summarizes the dominant themes identified in students’ reflections.

Table 4
Emerging Themes in Students’ Reflective Responses

Theme	Description	Sample Expression
Planning awareness	Anticipating content and structure	“I thought about what I wanted to say first.”
Monitoring awareness	Noticing clarity while speaking	“I stopped and tried again because it was not clear.”
Reflective evaluation	Evaluating speaking after completion	“Next time I will explain more slowly.”

These results indicate that metacognitive awareness became increasingly embedded in students’ speaking practices over repeated sessions.

5. Discussion

This study demonstrates that primary students’ speaking practices can function as a viable medium for developing metacognitive regulation when reflective routines are consistently habituated through digital microlearning support. This finding extends prior metacognitive research, which has largely focused on reading and writing, by empirically positioning speaking as a domain for metacognitive development in primary education (Flavell, 1979; Whitebread et al., 2009; Goh & Liu, 2023).

In addressing the first research question, the findings suggest that metacognitive awareness in speaking was habituated through repetition and consistency rather than explicit instruction. The gradual emergence of planning, monitoring, and reflection behaviors aligns with habit formation theory, which emphasizes that stable routines embedded in meaningful contexts support internalization of cognitive regulation (Lally et al., 2010). For young learners, such implicit habituation is particularly salient, as metacognitive processes are more effectively developed through practice than through abstract explanation (Veenman et al., 2006). The speaking activities in this study provided repeated opportunities for learners to enact

reflective behaviors, allowing metacognitive awareness to become part of routine speaking practice rather than an externally imposed requirement.

Regarding the second research question, the observed behavioral indicators offer concrete evidence of metacognitive engagement in primary students' speaking. Pausing to plan before speaking, self-correction during speech, and reflective evaluation after speaking correspond to the core metacognitive processes of planning, monitoring, and evaluation identified in metacognitive theory (Flavell, 1979). The prominence of post-speaking reflection supports previous findings that reflection serves as an accessible entry point for metacognitive development, particularly in oral language learning where cognitive processes are otherwise difficult to observe (Zhang & Zhang, 2019). This finding also corroborates research suggesting that structured reflection enables learners to gradually assume responsibility for regulating their own learning (Andrade & Brookhart, 2020).

The third research question concerned students' responses to digital microlearning as a support for reflective speaking routines. The findings indicate that microlearning functioned as an effective scaffold by providing concise and consistent prompts that reinforced reflective behavior without increasing instructional complexity. This supports prior research showing that microlearning is most effective when used to support routine formation and cognitive focus rather than as a primary content-delivery mechanism (Leong et al., 2021). As illustrated in **Figure 1**, digital microlearning was integrated across three phases of speaking, before, during, and after speaking, reinforcing metacognitive engagement as a cyclical and habitual process rather than a one-time instructional event. The figure visually clarifies how microlearning operated as contextual support within speaking activities, emphasizing habituation through repetition rather than technological novelty.

Taken together, these findings contribute to the broader discourse on speaking pedagogy by challenging performance-oriented approaches that prioritize fluency and accuracy over reflective engagement (Goh & Burns, 2012; Mercer & Dawes, 2014). By embedding metacognitive routines into speaking through habituation and lightweight digital scaffolding, the study bridges the gap between theoretical calls for meaningful, self-regulated learning and the realities of everyday primary classroom practice. This perspective reframes speaking not merely as oral output but as a pedagogical site for cultivating metacognitive regulation from an early stage of language development.

6. Conclusion

This study concludes that metacognitive awareness in primary students' speaking can be effectively habituated through consistent reflective routines supported by digital microlearning, rather than through explicit metacognitive instruction. Quantitative findings show a substantial increase in observable metacognitive speaking behaviors, with overall mean scores rising from 1.32 at baseline to 2.64 at the end of the intervention, alongside marked growth in planning, monitoring, and post-speaking reflection. These numerical trends are reinforced by qualitative

evidence indicating a shift from evaluative and surface-level comments toward process-oriented reflections, as well as by behavioral changes such as increased self-correction and voluntary reflection, which rose from 14 percent to 71 percent across sessions. Students' responses further indicate that digital microlearning functioned as an effective scaffold that supported consistency and recall of reflective routines without adding instructional burden, confirming its role as a contextual support rather than a technological focus. Taken together, the findings demonstrate that speaking can serve as a viable medium for developing metacognitive regulation in primary education when reflective practices are embedded as habits through repetition and supported by lightweight digital tools, thereby addressing the gap between meaningful learning ideals and routine classroom speaking practices.

7. Suggestion

(In this section, provide recommendations that supporting reported results during the study)

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Declarations

Author Contributions. (For example; A.B.: Literature review, conceptualization. C.D.: methodology, data analysis. E.F.: review-editing and writing, original manuscript preparation. All authors have read and approved the published on the final version of the article)

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Appendix I. (as applicable)

Appendix II. (as applicable)

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